Project Synopsis: Automated Dimension Ballooning

Introduction:

In industrial manufacturing, quality assurance relies on strict adherence to engineering drawings. The manual process of "ballooning" (assigning unique sequential identifiers to dimensions) and transferring this data to Excel inspection sheets is highly time-consuming and prone to human error. This project develops an interactive, web-based software tool that digitalizes and automates this workflow. By providing a dynamic PDF canvas, dual-mode data extraction (vector parsing and manual fallback), real-time tolerance validation, and collaborative multi-user inspection, the system significantly reduces preparation time and ensures 100% traceability between physical parts and digital reports.

Problem Statement:
Quality inspectors currently waste hours manually drawing balloons on 2D engineering PDFs and typing nominal values and tolerances into Excel sheets.

- **Inefficiency:** Transferring data manually from dense, complex diagrams to spreadsheets creates bottlenecks in production.
- **Software Limitations:** Existing prototype solutions struggle with spatial awareness, resulting in overlapping balloons that obscure critical drawing dimensions.
- **Lack of Collaboration:** Inspection is currently a solitary, siloed task; multiple inspectors cannot easily work on the same massive diagram simultaneously without creating conflicting Excel files.

Objective:

1. To develop an interactive drag-and-drop UI allowing users to dynamically place inspection balloons on PDF drawings without obscuring critical engineering text.
2. To implement a dual-mode data entry system: auto-extracting vector text when available, with a graceful manual fallback for static/scanned images.
3. To build an automated validation engine that instantly calculates whether actual physical measurements fall within nominal tolerance limits, providing color-coded feedback (Green/Yellow/Red).
4. To enable live, WebSocket-driven multi-user collaboration, allowing several technicians to inspect the same drawing simultaneously.
5. To automate the generation of FAIR/PPAP compliant Excel inspection sheets and final marked-up PDFs.

Scope of the Project:

- **In Scope:**
  - Development of a React-based frontend with an interactive Canvas/PDF viewer.
  - Implementation of the automated sequential numbering and bounding-box collision avoidance logic.
  - Backend logic for tolerance math (Nominal ± Tolerance).
  - Real-time multi-user syncing via WebSockets.
  - Exporting color-coded PDF drawings and structured Excel files.
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- **Out of Scope:**
  - Fully autonomous GenAI pre-extraction of all GD&T symbols without human input.
  - Direct integration into external ERP/SAP systems.
  - Deep CAD file (DWG/DXF) parsing.

Literature Review:
Current quality inspection workflows heavily rely on physical paper prints and standard desktop PDF editors, paired with manual Excel data entry. While some early-stage automation tools exist, they often fail when faced with highly dense drawings, placing balloons blindly and causing overlapping text issues (as seen in early foundry industry prototypes). Furthermore, existing systems operate completely offline or as single-user desktop applications, lacking the collaborative, cloud-based capabilities necessary for modern, distributed shop-floor teams.

Methodology:
The system utilizes a microservices architecture to balance UI interactivity with heavy document processing:

- **Frontend Canvas:** Uses React.js integrated with PDF.js and Fabric.js to render the engineering drawing. This layer handles the drag-and-drop balloon placement and coordinate mapping.
- **Backend & State Management:** A Node.js application manages the active inspection state, tracking balloon coordinates, nominal values, and physical measurements. WebSockets facilitate live updates across multiple connected devices.
- **Document Processing Engine:** A Python microservice (utilizing PyMuPDF) handles the final extraction of vector data and the heavy rendering required to bake the final color-coded balloons and leader lines into the output PDF.
- **Database:** PostgreSQL is used to store relational data between drawings, inspection batches, and discrete measurement logs.

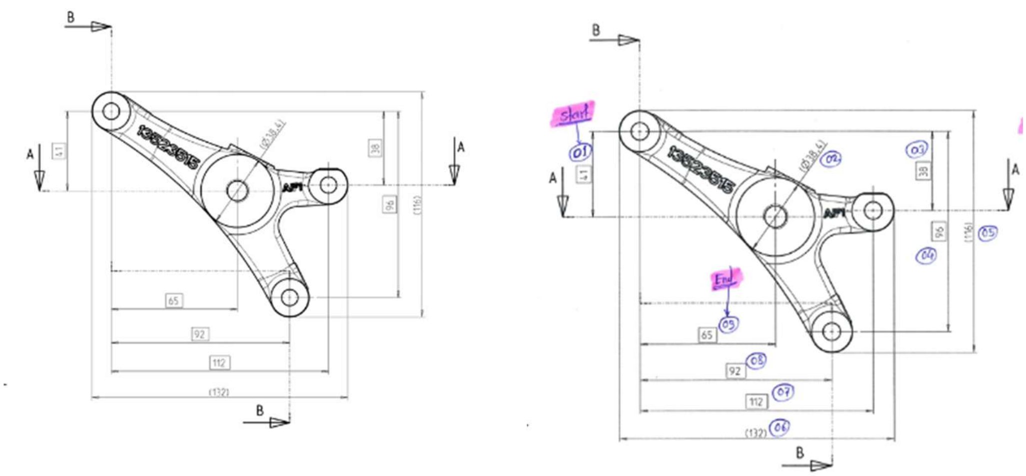

Fig. 01: Daigram Before Ballooning vs After

Modules of the Project:

1. **Ingestion & Canvas Module:** Handles the uploading, rendering, and scaling of complex vector and static PDFs.
2. **Interactive Ballooning Module:** Manages the click-to-balloon logic, auto-numbering, and drag-and-drop spatial coordinate tracking.
3. **Data Extraction & Fallback Module:** Attempts to read underlying vector text, defaulting to a manual-entry dialog box when vector data is absent.
4. **Validation & Math Engine:** Compares entered actuals against calculated tolerance limits to assign Pass/Check/Fail statuses.
5. **Collaboration Engine (WebSocket):** Syncs balloon statuses and measurement entries in real-time across multiple active user sessions.
6. **Reporting & Export Module:** Compiles the database entries into structured Excel sheets and regenerates the PDF with permanent colored markup.

Hardware & Software Requirements:

- **Software (Frontend):** React.js, PDF.js, Fabric.js, Tailwind CSS.
- **Software (Backend):** Node.js (Express), Python (FastAPI, PyMuPDF, exceljs), Socket.io.
- **Database:** PostgreSQL.
- **Hardware:** Standard desktop/tablet for web browser access; no specialized heavy-compute hardware required on the client side.

Results:

- A minimum 70% reduction in the time required to prepare inspection sheets.
- 100% data traceability linking physical inspection results to specific visual nodes on the engineering drawing.
- Elimination of overlapping UI elements, resulting in clear, legible final PDF reports.
- A seamless multi-user inspection environment that prevents duplicate work on the shop floor.

	Layout Inspection Report		F/AFPL-QAS/12 Rev No : 01 Rev Date : 11.11.2023
Customer : Liebherr Machining, Bulle	**Drawing No :** 94050440201	**Article No :** 13523526	**Date : 11.11.2023**
Part Name : Console - Machining	**Drawing Index : 005**	**Rev No : 05**	

Sr. No.	Parameters	Dimension	Class	Checking Method	Observations 001	Observations 002	Observations 003	Remarks
	Top View							
1	Tapping Hole Center Distance	41.0 ± 0.20		CMM / Relation Gauge	40.953	41.011	40.878	OK
2	Ref Casting Outer Dia (at parting line)	Ø 38.40 ± 0.65		Digital Vernier	38.36	38.28	38.35	OK
3	Hole Center Distance	38.0 ± 0.15		CMM / Relation Gauge	37.983	37.986	37.984	OK
4	Hole Center Distance	96.0 ± 0.15		CMM / Relation Gauge	95.976	95.977	95.968	OK
5	Reference End to End Dimension	116.0 ± 0.90		CMM of Casting Samples	AF1 - 116.403, AF2 - 116.599, AF3 - 116.692, AF4 - 116.382			OK
6	Reference End to End Dimension	132.0 ± 0.90		CMM of Casting Samples	AF1 - 132.327, AF2 - 132.563, AF3 - 132.479, AF4 - 132.669			OK
7	Hole Center Distance	112.0 ± 0.15		CMM / Relation Gauge	111.950	111.958	111.952	OK
8	Hole Center Distance	92.0 ± 0.15		CMM / Relation Gauge	91.980	91.918	91.973	OK
9	Tapping Hole Center Distance	65.0 ± 0.20		CMM / Relation Gauge	65.002	64.880	64.908	OK

Applications:

- **Shop-Floor Quality Assurance:** Enabling mechanics and technicians to log measurements dynamically on tablets while at their workstations.
- **FAIR/PPAP Documentation:** Instantly generating the required compliance paperwork for industrial manufacturing clients.
- **Supplier Audits:** Allowing external suppliers and internal engineers to collaborate on the same dimension sheet remotely.

Future Scope:

- **Automated Revision Migration:** Developing a vector-diffing engine to automatically port existing balloons to newer drawing revisions (e.g., Rev A to Rev B).
- **SPC Analytics Dashboard:** Tracking historical defect rates to predict tool wear and prevent out-of-tolerance failures before they occur.
- **GenAI Pre-Extraction:** Utilizing advanced vision models to parse complex GD&T symbols and fully pre-populate the nominal limits upon document upload.

Conclusion:

The Automated Dimension Ballooning Tool addresses a critical bottleneck in manufacturing quality control. By blending a highly interactive, user-friendly frontend with a robust mathematical validation backend, the system bridges the gap between static engineering drawings and actionable inspection data. The inclusion of real-time collaboration and dual-mode data entry ensures the tool is highly resilient and adaptable to the practical realities of the modern shop floor, significantly reducing administrative overhead and improving overall production quality.